

Solution Requirements

Date	15 March 2026
Team ID	6aba25588343c5cff063937b
Project Name	Comic Craft AI
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users sign up and log in with email and password or Google sign-in. Passwords are stored hashed; sessions use secure tokens, with logout and password reset.	High
2	Authorization levels	Guest: view the public gallery. Creator: generate, edit, export and share own comics. Admin: manage users, moderate gallery content and monitor usage.	Medium
3	External interfaces	Connects to an AI image-generation model API and a language model API for storyboards, plus cloud storage for	High

		images and an optional translation API for dialogue.	
4	Transactions processing	Each request (story prompt, panel generation, edit, export) runs as a tracked job with a status. Results are saved to the Comic Datastore, and failed jobs can be retried without losing data.	Medium
5	Reporting	Creators see a project history and can download PDF/PNG files. An admin dashboard shows usage, generation counts and error reports.	Low
6	Business rules, etc.	A story script and art style are required to generate a comic. Characters must stay consistent across all panels. Free users get a daily generation limit.	High
7	Compliance to laws or regulations	Follow data privacy laws (GDPR, India's DPDP Act). Block unsafe content, respect copyright and label AI-generated comics. Users can delete their account and data.	Medium
8	<i>Other</i>	Multilingual dialogue with automatic translation, and editing of speech bubbles and panel layout before export.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	Comic panels are generated quickly and the editor responds smoothly.	Page loads in under 3 seconds; one panel generated in under 15 seconds
2	Scalability	The system handles growth in users and stored comics by scaling cloud services.	Supports 1,000 concurrent users with auto-scaling storage
3	Security & Data Privacy	User accounts, scripts and images are protected with encryption and secure access, with regular vulnerability checks.	HTTPS/TLS in transit; AES-256 at rest; hashed passwords
4	Reliability & Availability	The service stays available; failed generation jobs retry automatically and data is backed up.	99.5% uptime per month; daily backups
5	Usability & Accessibility	A simple, guided interface for non-artists on desktop and mobile, with accessible design.	Complies with WCAG 2.1 AA; first comic in under 5 minutes
6	<i>Other</i>	Maintainability and portability: modular code and cloud-independent deployment.	Modular design; works on all major browsers